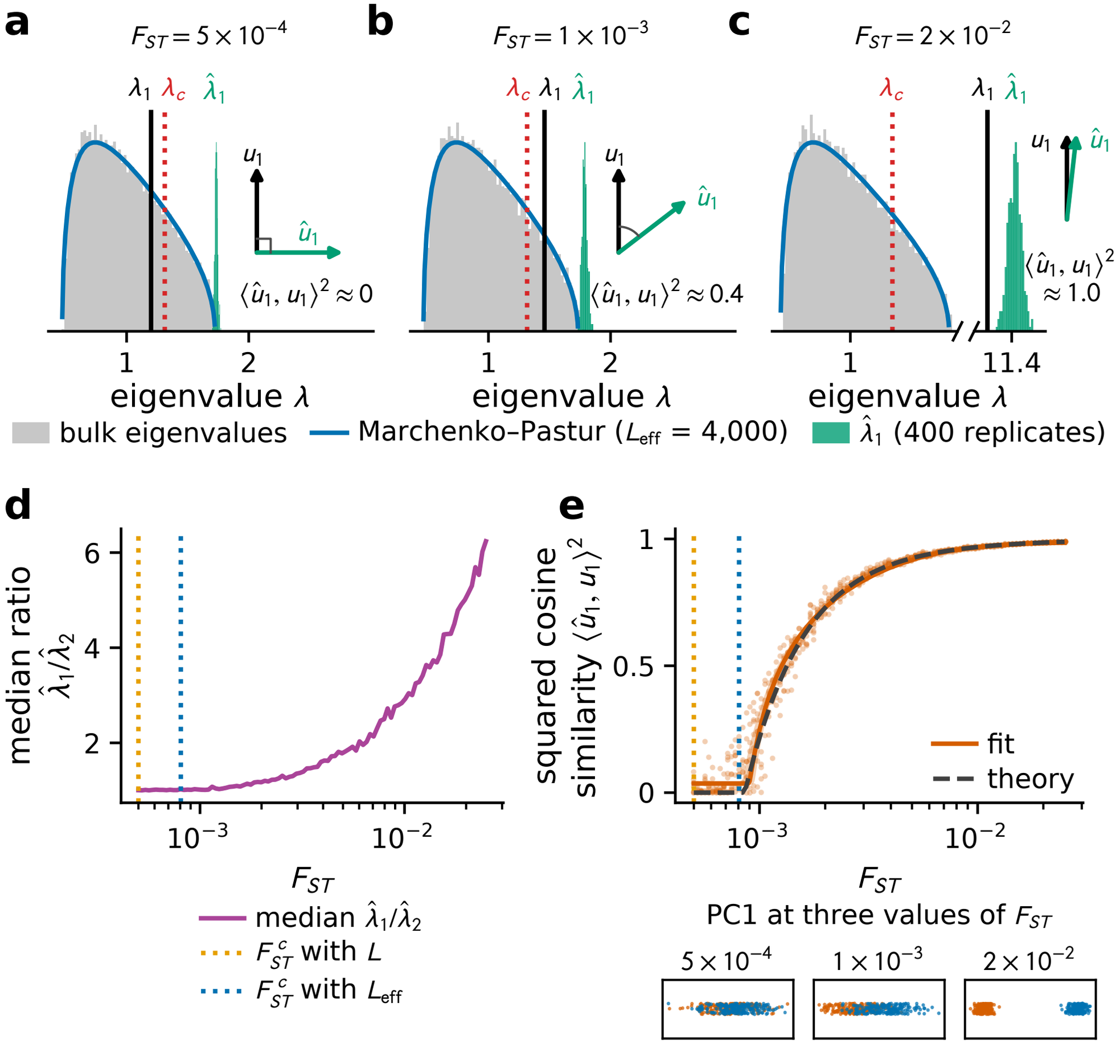

a
$F_{ST} = 5 \times 10^{-4}$
λ_1 λ_c $\hat{\lambda}_1$
u_1
$\hat{u}_1$
$\langle \hat{u}_1, u_1 \rangle^2 \approx 0$
1
2
eigenvalue λ
b
$F_{ST} = 1 \times 10^{-3}$
λ_c λ_1 $\hat{\lambda}_1$
u_1
$\hat{u}_1$
$\langle \hat{u}_1, u_1 \rangle^2 \approx 0.4$
1
2
eigenvalue λ
c
$F_{ST} = 2 \times 10^{-2}$
λ_c
λ_1 $\hat{\lambda}_1$
u_1
$\hat{u}_1$
$\langle \hat{u}_1, u_1 \rangle^2 \approx 1.0$
1
11.4
eigenvalue λ
bulk eigenvalues
Marchenko–Pastur ($L_{\text{eff}} = 4{,}000$)
$\hat{\lambda}_1$ (400 replicates)
d
median ratio $\hat{\lambda}_1/\hat{\lambda}_2$
6
4
2
10^{-3}
10^{-2}
F_{ST}
median $\hat{\lambda}_1/\hat{\lambda}_2$
F_{ST}^c with L
F_{ST}^c with L_{eff}
e
squared cosine similarity $\langle \hat{u}_1, u_1 \rangle^2$
1
0.5
0
10^{-3}
10^{-2}
F_{ST}
fit
theory
PC1 at three values of F_{ST}
5×10^{-4}
1×10^{-3}
2×10^{-2}